Lesson 10.1 - Hypothesis Testing - Null and Alternative Hypotheses

> **Video:** Hypothesis Testing - Null and Alternative Hypotheses by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

The two hypotheses

	Null hypothesis H_0	Alternative hypothesis H_a (or H_1)
What it is	The **status quo**: the currently accepted value or claim	The **challenger**: what the researcher is trying to show
Sign	Contains the **equality**: $=, \leq, \geq$	**Strict** inequality: $\neq, <, >$

The researcher tries to gather enough evidence to **reject** H_0 in favor of H_a.

> **Careful wording:** if the evidence isn't strong enough, we **fail to reject** H_0. That doesn't prove H_0 is true, or even that it is "likely" true. It only means the data didn't rule it out.

Steps

1. **Mean or proportion?**
   - "average", "mean" → population mean μ
   - "percent", "proportion", "fraction" → population proportion p, written as a decimal (80% → 0.80)
2. **Find the accepted value** (the status quo) and write H_0 with the equality.
3. **Find what the researcher suspects** and write H_a with the strict inequality. Together, H_0 and H_a cover every possibility.

Wording	H_0	H_a
"is different from", "is not"	$=$	$\neq$
"at least", "no less than"	$\geq$	$<$
"at most", "no more than"	$\leq$	$>$

Why the signs go this way

1. H_0 gets the equality because the math needs one exact number. A test starts by assuming H_0 is true, then asks: "If that were true, how surprising is my sample?" To measure "how surprising" you need one specific value to plug into the formula.

- $\mu = 10$ gives you 10, so you can calculate.
- $\mu < 10$ doesn't. Is it 9.9? 7? 2? You can't compute anything from that.

So the exact number (the equality) always lives in H_0. Even when H_0 is $\geq$ or $\leq$, the calculation uses the boundary value, the "=" part.

2. H_0 gets the benefit of the doubt. Think of a courtroom:

- H_0 = **innocent** (the default, believed until proven otherwise)
- H_a = **guilty** (what the prosecutor has to prove)

The researcher is the prosecutor. The burden of proof is on H_a, so H_a gets the strict claim ($\neq$, $<$, $>$).

3. Together they cover every possibility, with no overlap. That's why they come in fixed pairs:

H_0	H_a	Together
$=$	$\neq$	equal or not equal → everything
$\geq$	$<$	at least, or below → everything
$\leq$	$>$	at most, or above → everything

Example. A company says its battery lasts **at least 10 hours**. You suspect it's less.

$$H_0 : \mu \geq 10 \qquad H_a : \mu < 10$$

H_0 is the company's claim (status quo, includes the "="). H_a is your suspicion (strict).

What "fail to reject" means

"Fail to reject H_0" = "not enough evidence to say H_0 is wrong." It does NOT mean "H_0 is right."

- **Courtroom:** the jury says "not guilty", never "innocent". Maybe the person really is innocent, or maybe the prosecutor just didn't have enough evidence. The verdict only says the case wasn't proven.
- **Lost keys:** you search your room for 30 seconds and don't find your keys. That doesn't prove they aren't there. You just didn't find them. Maybe you didn't look long enough, which in statistics is like having too small a sample.
- **Battery:** you test 5 batteries and get an average of 9.8 hours. That's a bit below 10, but with only 5 batteries it could easily be random luck. So you fail to reject H_0. You haven't proven the batteries last 10+ hours. You just can't call the company a liar yet.

Decision	Meaning
Reject H_0	Strong evidence. H_0 looks wrong, so we side with H_a.
Fail to reject H_0	Not enough evidence. We can't say H_0 is wrong. (Never say "accept H_0".)

Only two decisions: reject or fail to reject

There are only **two** possible decisions about H_0. "Accept H_0" is **not** a third option; it's just a wrong way of saying "fail to reject". A test can only collect evidence **against** H_0, never evidence **for** it.

The courtroom has the same two verdicts:

- **Guilty** = reject H_0
- **Not guilty** = fail to reject H_0

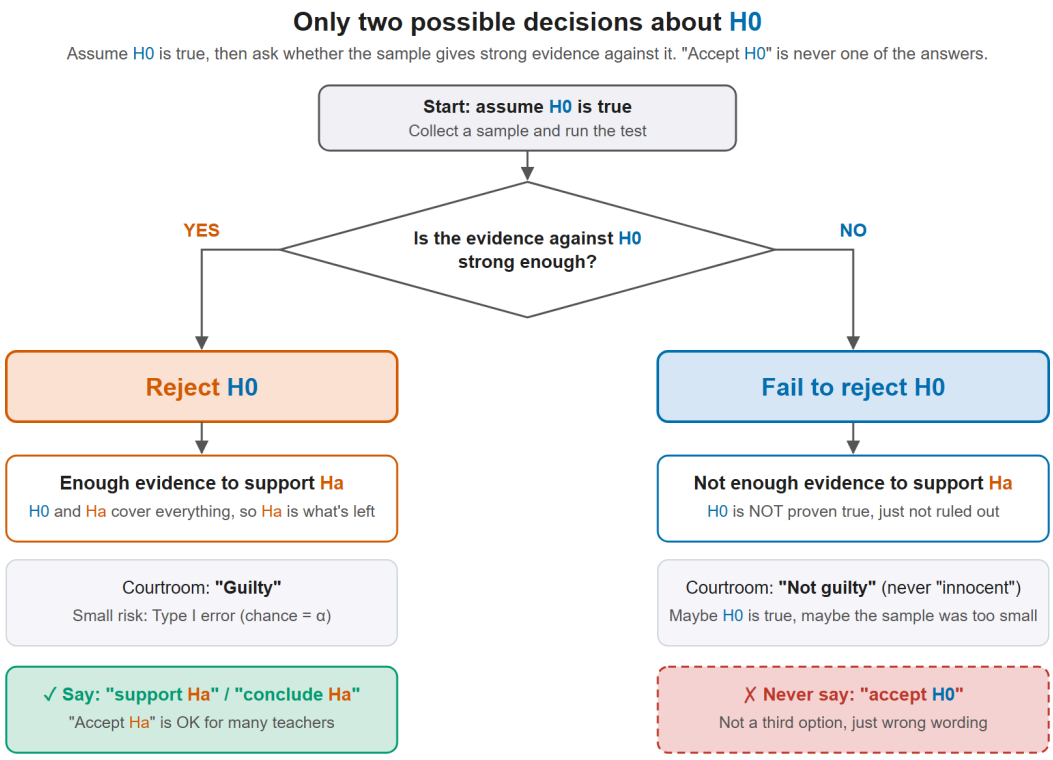
There's no "innocent" verdict, just as there's no "accept H_0".

Reject H_0 means you support H_a. H_0 and H_a cover every possibility with no overlap. If you have strong evidence that H_0 is wrong, the only option left is H_a. It doesn't work the other way: failing to reject H_0 says

nothing in favor of H_0.

Decision	What you can say about H_a
Reject H_0	Enough evidence to **support** H_a
Fail to reject H_0	Not enough evidence to support H_a (and H_0 is not proven true)

> **Small caution:** "support H_a" or "conclude H_a" is safer wording than "accept H_a", because rejecting H_0 isn't 100% proof. There's a small chance you rejected H_0 when it was actually true. That's a **Type I error**, and its chance is α (for example 5%). See Lesson 10.2.



Exam-friendly wording:

- ✅ "Reject H_0. There is sufficient evidence to support the claim that $\mu < 10$."
- ✅ "Fail to reject H_0. There is not enough evidence to support the claim that $\mu < 10$."
- ❌ "Accept H_0." (Some older textbooks use it, but most teachers mark it wrong.)

Examples

1. Calculator mass. Company XYZ's calculators average **450 g**. An engineer thinks the average is different and weighs 50 calculators.

$$H_0 : \mu = 450 \qquad H_a : \mu \neq 450$$

("Average" means a mean.)

2. Graduation rate. Teachers believe **at least 80%** of students graduate. A student disagrees.

$$H_0 : p \geq 0.80 \qquad H_a : p < 0.80$$

("Percent" means a proportion. "At least 80%" means 80% or more.)

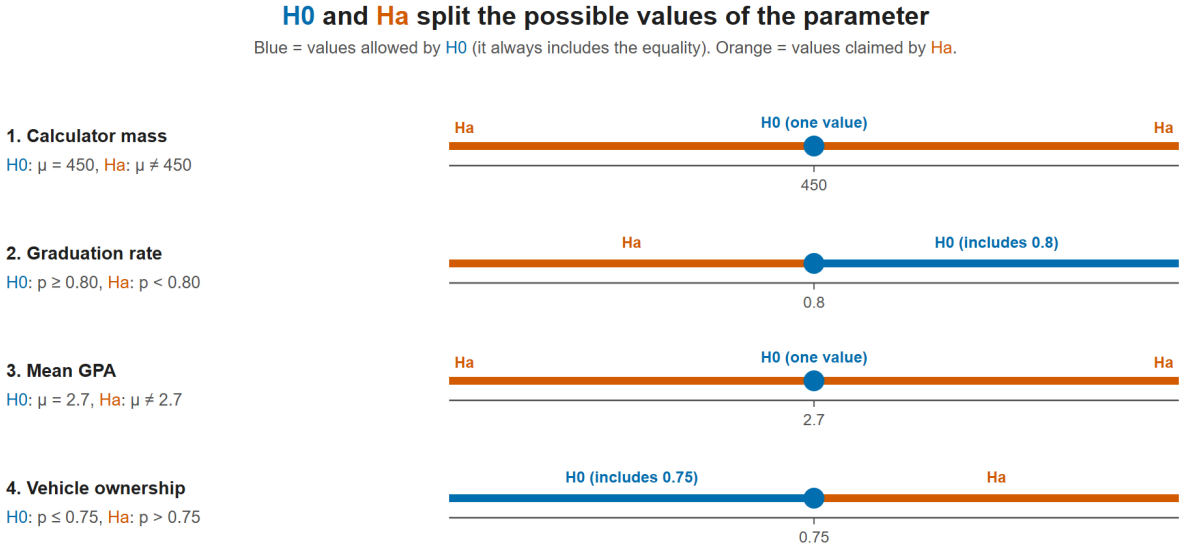
3. GPA. A teacher wants to test whether the mean GPA is **different from 2.7**. Here it is easier to start with H_a, the teacher's claim.

$$H_0 : \mu = 2.7 \qquad H_a : \mu \neq 2.7$$

4. Vehicle ownership. The share of residents who own a vehicle is **no more than 75%**. A researcher disagrees and surveys 100 residents.

$$H_0 : p \leq 0.75 \qquad H_a : p > 0.75$$

("No more than" means ≤. Surveying 100 residents gives the sample size, which is not part of either hypothesis.)



H_0 and H_a always cover every possible value between them. H_0 holds the boundary value (the equality). With "≠", H_0 is a single point and H_a is everything else.

Key takeaways

- H_0 = status quo with the equality ($=$, $\leq$, $\geq$). H_a = the claim being tested, with a strict sign ($\neq$, $<$, $>$).
- Hypotheses are about **population parameters** (μ or p), never about sample values like $\bar{x}$ or $\hat{p}$.
- Translate the wording carefully: "at least" → $\geq$, "no more than" → $\leq$, "different" → $\neq$.
- H_0 holds the equality because the test needs one exact value to calculate with, and H_0 gets the benefit of the doubt (like "innocent until proven guilty").
- "Fail to reject H_0" means "not proven wrong", not "proven right". Never say "accept H_0".
- Only two decisions exist: **reject** H_0 (→ support H_a) or **fail to reject** H_0 (→ not enough evidence for H_a).